NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

School of Computing

Department of Software Engineering

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MAP711S (Mobile Application Development)

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students at the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit to abide by the institution's policies and rules. The core activity of NUST is learning, and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST therefore does not condone any form of academic dishonesty, including plagiarism, cheating on tests and assessments, and other such practices. NUST requires students to always do their own assignments and to produce their own academic work, unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works, or inventions of someone else as if they were your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

COURSE INFORMATION

COURSE CODE AND TITLE:

MAP711S, Mobile Application Development

DEPARTMENT:

Software Engineering

PROGRAMME:

Bachelor of Computer Science: Software Development

CONTACT HOURS:

Contact hours 45, Self-study hours 60, Assessment hours 15

NQF LEVEL AND CREDIT:

NQF level 7, 12 credits

COURSE DESCRIPTION:

This course will equip students with the necessary skills and knowledge of advanced software design, development, testing, and management techniques. In addition, students should also be able to critique opportunities and challenges associated with developing software applications in a specific domain among Artificial Intelligence, blockchain, and Internet of Things (IoT).

PRE-REQUISITES:

Programming 2

COURSE EQUIVALENCIES:

None

COURSE DELIVERY METHODS:

The course will be facilitated through the following learning activities:

- Lectures - this will present the underpinning theories of the course and students are expected to deepen their knowledge and understanding of the covered subjects through guided self-study.
- Laboratory activities – this will be an avenue for the students to try out and acquire practical skills of theoretical knowledge gained. Laboratory materials will be prepared with a clear indication of what the student is expected to achieve, and the problem statement and guidelines on how to approach the problem.

The following communication tools will be used in this course:

E-learning Platform (Moodle), Online Server (ISNOTES), Cellphone (WhatsApp)

Course Format:

- 45 hours of face-to-face (or online via MS Teams) between students and lecturer
- 60 hours of out-of-class study time, to be used as follows:
  - 20 hours of laboratory (practical) session
  - 20 hours of reading prescribed material
  - 5 hours of sorting, completing and annotating own lecture notes
  - 10 hours of working on assignments
  - 5 hours of consultations with lecturer, laboratory session manager and fellow students
- 15 hours of assessments:
  - 5 hours of completing online quizzes
  - 5 hours of tests
  - 5 hours of class and project presentations

EFFECTIVE DATE:

Monday 9 February 2026

LECTURER INFORMATION

Lecturer's name: Simon Muchinenyika
Email: smuchinenyika@nust.na
Office phone: 2072054
Office location: No. 201, Polyheights
Office hours: 07h30-12h30
Consultation hours: Thursdays 17h00

STUDENT READINESS

Technology & Equipment Readiness:

This course will only utilise already existing NUST equipment. Students will use their own laptops for implementation and must ensure appropriate backup.

Student Commitments and Contact Times:

1. Attend all classes and make appropriate notes. The presentations used in class and the laboratory materials will be made available on Moodle or, if there is a glitch, on MS Teams. If a student misses a class, it is their responsibility to catch up on what was covered.
2. After every class and before the start of the next class, the student should sort, complete, and annotate their own lecture notes, and select further reading if any item was not understood on the first attempt.
3. Students must fully utilise their notational hours to revise, practice, and/or complete homework for the course.
4. Consult the recommended material to further broaden knowledge and understanding of the subject.

Course Resources:

Presentation slides, laboratory materials, and additional materials will be made available on e-learning (Moodle).

Prescribed Reading:

Phillips, B., Stewart, C. & Marsicano, K. (2017). *3rd Edition Android Programming.* The Big Nerd Ranch Guide, ISBN-10: 0134706072.

Recommended Reading:

1. Helal, S., Li, W. & Bose, R. (2012). *Mobile platforms and development environments*. Morgan & Claypool Publishers, ISBN-10: 1608458660.
2. Wargo, J. M. (2012). *PhoneGap essentials: Building cross-platform mobile apps*. Addison Wesley, ISBN-10: 0321814290.

STUDENT LEARNING

Learning Outcomes:

On completing the course, students will, through assessment activities, show evidence of their ability to:

1. Distinguish different mobile platforms by identifying their strengths and weaknesses for both the developer and the user.
2. Set up and use a development environment for a specific platform.
3. Design mobile application user interfaces (UIs) to match user requirements.
4. Demonstrate essential skills needed to develop mobile applications on a specific platform.
5. Develop a mobile application that uses a network service.

COURSE SCHEDULE:

Week Beginning	Topics	Tasks
9 February	**Introduction** Mobile Landscape, Choosing a Platform, Opportunities & Challenges	• *Teams & Moodle Registration* • *Install Android Studio* • *Jetpack Compose Lab*

16 February	**Android Application Architecture** App Composition, Architectural Principles, App Architecture	• *Activities, Content Providers, Services, Broadcasts* • *App Architecture Link*
23 February	**Language Fundamentals** Variables, Functions, Expressions, Objects, Abstraction, Inheritance	In-class Lab 1 • *Kotlin Lab Link* • *Lambda Expressions*
2 March	**Android Libraries** Android Packages, Jetpack, Compose, Google Play Services & SDK	• *Now in Android App GitHub Link* • *Data & UI Layers*
9 March	**Generics in Kotlin** Iteration, Operations & Transformation	Mastery Learning Quiz 1 • *Generics Lab* • *Code lab Link*
16 March	**Android Navigation** Styles, Layouts & Content, Behaviours & Patterns, Accessibility	In-class Lab 2 • *Jetpack Navigation 3. GitHub Link*
23 March	**Test** Group Project Discussions	Test Group Project Release
30 March	**Mid-Semester Break**	**Break**
7 April	**Kotlin Collections** Iteration, Operations & Transformation	In-class Lab 3 • *Taxonomy* • *Using Collections Codelab*
13 April	**Modularisation in Android** Cohesion/Coupling Types of Modules	• *Multi Module Sample GitHub Link* • *Dependency Inversion*
20 April	**Functional Programming** Lazy computations Operator overloading	Mastery Learning Quiz 2 • *Higher-order Functions* • *Code-lab Link*
27 April	**Android Testing** Types of Tests, Testing Strategies, Testing APIs	In-class Lab 4 • *Android Testing Samples* GitHub Link
5 May	**Parallel & Concurrent Programming** Parallelism Vs Concurrency Creating Threads	Supplementary Test • *Synchronisation* • *Finalizing Group Project*
11 May	**Group Project Presentation**	Group Project Presentations

NOTE: Test and assignment dates are subject to change based on the needs of the students at the lecturer's prerogative. Students will be notified ahead of time of any changes.

ASSESSMENT AND EVALUATION:

The semester mark is determined by continuous evaluation made up of Test and individual assignments. The final course mark is the sum of the quizzes, test and individual assignment.

Assessment	Weight
In class Labs	20%
Test	25%

Group Project	40%
Quizzes	15%
Total:	**100%**

Minimum pass requirement for this course: 50%

COURSE POLICIES

General Academic Policies:

It is the student's responsibility to be familiar with and adhere to NUST's Policies. These policies can be found in the NUST Prospectus or online at www.nust.na/prospectus.

Supplementary Policies:

- If a student misses an in-class lab for a valid, provable reason, the remaining labs will be averaged instead of averaging by four. It is mandatory, though, to attempt at least two labs; otherwise, the student will get a zero.
- The supplementary test is open to those who missed the Test or want to improve their Test marks. However, for those who plan to improve their marks, the supplementary test mark is final, whether higher or lower than the Test mark.
- Lastly, there are no supplementary assessments for quizzes or projects, as the assessments remain open for a long period.

All late submissions will not be accepted after the deadline and thus will be marked as 0%.

DATE REVISED: 7 February 2026

FAILURE TO PAY FEES:

A student who fails to pay his/her fees may have their results withheld until all outstanding fees are paid in full.

IMPORTANT STUDENT SERVICES AT NUST

There are a variety of services that you can use at the NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development – Department: Student Services
- Writing Centre and student academic problems –Teaching and Learning Unit (CTL)
- Campus Health and Wellness Centre (CHWC) - Student Services'/ NUST Clinic

AUTHORISATION:

This course is authorised for use by:

09/02/2026

_____ _____

Head of Department Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students in the course, detached from the course outline and kept on record in the department)

I, (insert name and student number) _____, hereby acknowledge that I have received this course outline for MAP711S, and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements set out in this course outline.

_____ _____

Signature of Student Date